

Sandhya Saravanan

✉ ssarav13@ucsc.edu
📁 [sandhyasaravanan99.github.io](https://github.com/sandhyasaravanan99)

Education

2025– **University of California, Santa Cruz**

Present PHD COMPUTER SCIENCE

CGPA: 4/4

Advisor: Dr. Ioannis Demertzis

Research Areas: Multi-Party Computation, Blockchain Security, Searchable Encryption

Courses: Cryptography, Network Security, Private Computation on Encrypted Data, Database Theory

2017–2021 **National Institute of Technology, Tiruchirappalli**

B.TECH COMPUTER SCIENCE

CGPA: 8.66/10

Courses: Cryptography, Data Structures and Algorithms, Database Management, Operating Systems, Networks, Automata and Formal Languages

Publications

- 1 Divyanshu Bhardwaj, Sandhya Saravanan, Nishanth Chandran, Divya Gupta. “**Securely Training Decision Trees Efficiently**”. 31st ACM Conference on Computer and Communications Security (CCS)

[Paper ↗](#) , [ACM CCS 2024](#)

Experience

2023-2025 **Microsoft Research, Bangalore**

RESEARCH FELLOW

Advisors: Dr. Divya Gupta, Dr. Nishanth Chandran

Working on applications of Secure Multiparty Computation (SMPC) to enable privacy preserving machine learning and database operations as a part of **EzPC**, the Cryptography team.

2021–2023 **Microsoft, India**

SOFTWARE DEVELOPER

Backend development of actionable messages and connectors integrated into Outlook and Teams.

Summer 2020 **SOFTWARE DEVELOPER INTERN**

Built a Power BI dashboard to enable cost optimization on Azure using Azure Databricks.

2020–2021 **Indian Institute of Science, Bangalore**

RESEARCH INTERN, CrIS Lab

Implemented Secure Multiparty Computation (SMPC) protocols from [Astra](#) in C++ using primitives from [ABY3](#). Analyzed and benchmarked custom SMPC friendly sigmoid functions for logistic regression and simple neural networks against [MP-SPDZ](#)

Honors and Awards

2022 Runner up in **Cloud CTF**, an Azure Cloud ethical hacking contest, Microsoft.

2022 Top 5 in **Contoso Skate and Contoso Financial CTF**, a web hacking contest, Microsoft.

2020 Finalist in **Smart India Hackathon**, a nationwide government-funded hackathon

2018 Top 10 in **InCTF**, India's first Attack Defence Capture the Flag.

2013-17 Top 250 in India to qualify for **Indian National Olympiad of Informatics (INOI)**.

2016 Top 300 in India to qualify for **Indian National Mathematical Olympiad (INMO)**.

Technical Skills

Blockchains, Searchable Encryption, Multi-Party Computation
MP-SPDZ, Linux, Git

Research
Tools and Systems

Academic Service and Extracurricular Activities

- **Teaching Assistant**, UCSC: Algorithmic Foundations of Cryptography
- **Women Inclusivity Network**, NIT Trichy: Co-founded a community of female students and alumni of NIT Trichy that aims to improve the representation of women in STEM fields.
- **Teach for India**: As a volunteer, developed workgen.org, an educational tool to automate question paper generation, with the goal of minimizing the gap in quality of teachers between cities and rural areas.
- Actively contributed to promoting coding culture on-campus by organizing hackathons, CTFs, programming contests and workshops for the student community as a member of Delta Force, NIT Trichy.